



Cluster : Hardware Security

Problem Statement title : HSM Tampering Detection System

Description:

Design and implement a real time monitoring system that can detect, log, and respond to tampering attempts on Hardware Security Modules (HSMs). The solution should focus on practical, demonstrable tamper scenarios such as voltage glitches, enclosure opening, fault injections, and sensor triggers.

The system must securely log tamper events, generate real time alerts, and produce compliance reports that align with international benchmarks (e.g., FIPS 140-3 Level 3+). While achieving full FIPS compliance may be beyond the challenge timeframe, participants are expected to address a meaningful subset of tamper detection features (e.g., side-channel aggregation, fault attack countermeasures, secure log storage) and validate their approach against simulated tamper events.

Exact Deliverables :

- Prototype tamper detection system with continuous telemetry and alerting.
- Side channel and sensor aggregation (voltage, enclosure switches, etc.) with secure logging.
- Test rig that validates against FIPS-140 tamper event requirements.
- Monitoring module with compliance ready reporting.

Milestones, Evolution Parameters:

Phase 1: Submission of proposed approaches and a baseline prototype on a simulated HSM, along with a detailed report.

Phase 2: Depending upon the reports of phase 1, shortlisted teams develop enhanced prototypes on low cost hardware platforms (e.g., ARM Juno Board, Raspberry Pi, or other OP-TEE supported boards).

Phase 3: Final demos of working solutions, evaluated against key FIPS 140-3 Level 3+ principles (tamper detection, evidence, response, and secure logging).

Additional Information:

- Teams should prioritize modular design so the system works across multiple HSM vendors.
- Consider secure storage for tamper logs (append-only, cryptographically signed).
- Solutions should support integration into enterprise SIEM dashboards.